TrES-2b

R-1391

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-2_190612` · created 2026-08-01T02:58:23+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458646.85921 +/- 0.00065 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.119 +/- 0.016
Transit depth (Rp/Rs)^2	1.41 +/- 0.38 [%]
Orbital Inclination (inc)	88.8 +/- 1.48
Airmass coefficient 1 (a1)	0.6368 +/- 0.0073
Airmass coefficient 2 (a2)	0.418 +/- 0.064
Scatter in the residuals of the lightcurve fit is	1.15 %
Best Comparison Star	#1 - [91, 224]
Optimal Aperture	4.11
Optimal Annulus	8.19
Transit Duration (day)	0.1084 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.119 ± 0.016 vs 0.1254 (deviation 0.29σ)
Duration fit vs published	0.1084 d vs 0.0746 d (deviation 5.26σ)
Mid-time O–C	-0.7 min
Depth SNR	5.4
Residual scatter	1.15 %

Light curve

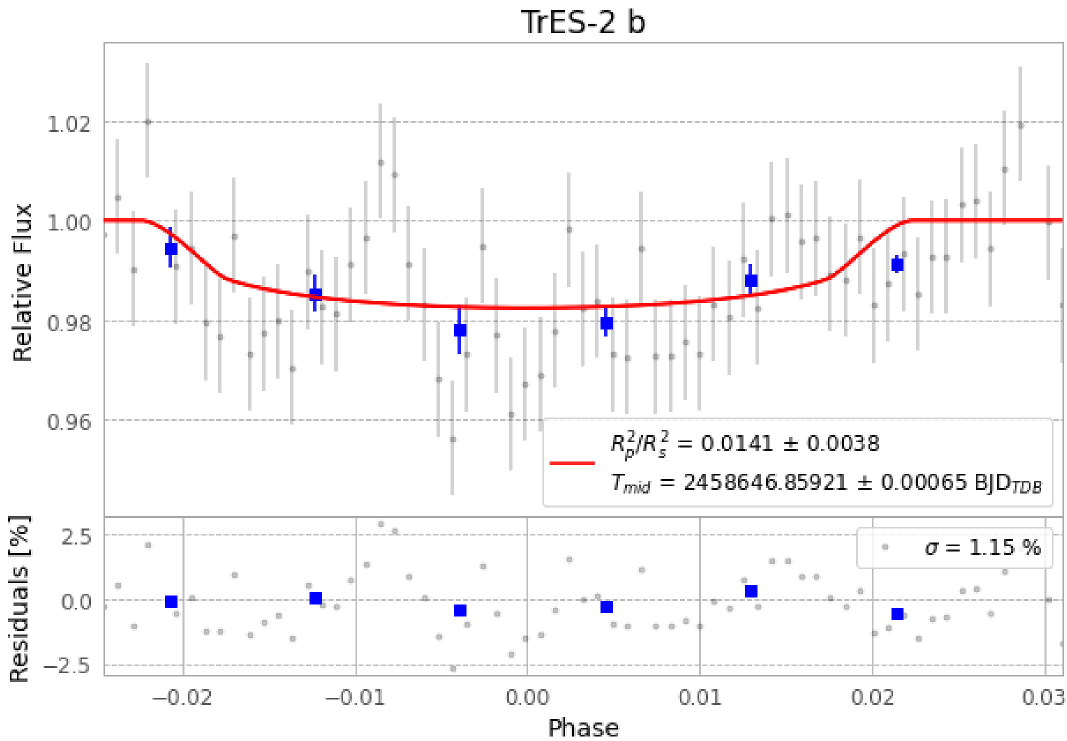

Figure 1 — FinalLightCurve_TrES-2 b_2019-06-12.png (SHA-256 f8077c13ca7c27c7…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [397, 180], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 a397849033a93f3f…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

67 FITS frames (44 MB), `TRES-2190612070612.FITS` → `TRES-2190612102412.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-190612.log (433bf97635bb…), FinalLightCurve_TrES-2 b_2019-06-12.png (f8077c13ca7c…), FinalLightCurve_TrES-2 b_2019-06-12.csv (f0e38294e829…)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-08-01 02:58Z.